

Differentiation of Trigonometric Function

1. Differentiate with respect to x :

- a. $\sin x$
- b. $\cos x$
- c. $\tan x$
- d. $\sin 2x$
- e. $2 \cos x$
- f. $2 \tan 2x$
- g. $\sin 2\pi x$
- h. $\tan \frac{\pi}{2} x$
- i. $3 \sin x + \cos 5x$
- j. $4 \sin \pi x + 3 \cos \pi x$
- k. $\cos(5x + 4)$
- l. $7 \sin(2 - 3x)$
- m. $10 \tan(10 - x)$
- n. $\frac{1}{\pi} \tan\left(\frac{\pi}{2} - 2\pi x\right)$
- o. $6 \sin\left(\frac{x+1}{2}\right)$
- p. $15 \cos\left(\frac{3-2x}{5}\right)$

2. Differentiate the following functions

- a. $y = \sin(x^2)$
- b. $y = \cos(x^3 + 1)$
- c. $y = \sin\left(\frac{1}{x}\right)$
- d. $y = \tan \sqrt{x}$
- e. $y = x \sin x$
- f. $y = x^2 \cos 2x$
- g. $y = \cos^2 x$
- h. $y = \sin^3 x$
- i. $y = \frac{1}{1+\sin x}$
- j. $y = \frac{\sin x}{1+\cos x}$
- k. $y = \frac{1-\sin x}{\cos x}$
- l. $y = \frac{\cos x}{\cos x + \sin x}$
- m. $y = \ln(\cos x)$
- n. $y = e^{\tan x}$

3. Sketch $y = \cos x$ for $-3\pi \leq x \leq 3\pi$. Find y' , y'' , y''' and y'''' , and sketch them underneath the first graph. What geometrical interpretations can be given of the facts that:

- a. $y'' = -y$?
- b. $y'''' = y$?

4. Find $f'(x)$, given:

- a. $f(x) = e^{\sin 2x}$
- b. $f(x) = \sin(e^{2x})$
- c. $f(x) = \sqrt{\sin 4x}$
- d. $f(x) = \ln(\sin 4x)$
- e. $f(x) = (1 - \cos 3x)^3$
- f. $f(x) = \sin 2x \sin 4x$
- g. $f(x) = \cos^5 3x$
- h. $f(x) = \frac{1}{3+4 \cos 5x}$
- i. $f(x) = \tan^3(5x - 4)$
- j. $f(x) = \sin \sqrt{x^2 + 1}$

5. Given that $y = e^{-x}(\cos 2x + \sin 2x)$, show that $y'' + 2y' + 5y = 0$

6. a. Express x° (that is, x degrees) in radians.

b. Hence find:

- i. $\frac{d}{dx}(\sin x^\circ)$
- ii. $\frac{d}{dx}(\tan(x^\circ + 45^\circ))$
- iii. $\frac{d}{dx}(\cos 2x^\circ)$

7. a. By writing $\sec x$ as $(\cos x)^{-1}$, show that

$$\frac{d}{dx}(\sec x) = \sec x \tan x.$$

b. Similarly, show that

- i. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
- ii. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

8. Given that $y = \frac{1}{3} \tan^3 x - \tan x + x$, show that $\frac{dy}{dx} = \tan^4 x$.

9. a. Copy and complete: $\log_b \left(\frac{P}{Q} \right) = \dots$

b. If $f(x) = \ln \left(\frac{1+\sin x}{\cos x} \right)$, show that

$$f'(x) = \sec x$$

10. a. Show that $\frac{1}{2}(\sin(m+n)x + \sin(m-n)x) = \sin mx \cos nx$.

b. Hence, without using the product rule, find the derivative of $\sin mx \cos nx$.

c. Simplify $\frac{1}{2}(\cos(m+n)x + \cos(m-n)x)$, and hence differentiate $\cos mx \cos nx$.

11. a. If $y = \sin x$, prove

i. $\frac{dy}{dx} = \sin \left(\frac{\pi}{2} + x \right)$

ii. $\frac{d^2y}{dx^2} = \sin(\pi + x)$

iii. $\frac{d^3y}{dx^3} = \sin \left(\frac{3\pi}{2} + x \right)$

b. Deduce an expression for $\frac{d^ny}{dx^n}$.

12. a. If $y = \ln(\tan 2x)$, show that $\frac{dy}{dx} =$

$$2 \sec 2x \operatorname{cosec} 2x.$$

b. Show that $\frac{d}{dx} \left(\frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x \right) =$

$$\sin^4 x \cos^3 x.$$

c. Show that $\frac{d}{dx} \left(\frac{\cos x + \sin x}{\cos x - \sin x} \right) = \sec^2 \left(\frac{\pi}{4} + x \right)$.

d. i. Show that $\cos 2x = \cos^2 x - \sin^2 x$ by using the formula for $\cos(A+B)$.

ii. Hence show that

$$\frac{d}{dx} \left(\frac{1+\cos 2x}{1-\cos 2x} \right) = -2 \cot x \operatorname{cosec}^2 x.$$

e. Show that $\frac{d}{dx} \left(\ln \frac{\sqrt{2}-\cos x}{\sqrt{2}+\cos x} \right) = \frac{2\sqrt{2} \sin x}{1+\sin^2 x}$

13. Given that parametric equation $x = e^t \cos t$,

$y = e^t \sin t$, show that $\frac{dy}{dx} = \tan \left(t + \frac{\pi}{4} \right)$.

Answer

1. a. $\cos x$; b. $-\sin x$; c. $\sec^2 x$; d. $2 \cos 2x$; e. $-2 \sin x$; f. $4 \sec^2 2x$; g. $2\pi \cos 2\pi x$; h. $\frac{\pi}{2} \sec^2 \frac{\pi}{2} x$; i. $3 \cos x - 5 \sin 5x$; j. $4\pi \cos \pi x - 3\pi \sin \pi x$; k. $-5 \sin(5x+4)$; l. $-21 \cos(2-3x)$; m. $-10 \sec^2(10-x)$; n. $-2 \sec^2 \left(\frac{\pi}{2} - 2\pi x \right)$; o. $3 \cos \left(\frac{x+1}{2} \right)$; p. $6 \sin \left(\frac{3-2x}{5} \right)$
2. a. $2x \cos(x^2)$; b. $-3x^2 \sin(x^3+1)$; c. $-\frac{1}{x^2} \cos \left(\frac{1}{x} \right)$; d. $\frac{1}{2\sqrt{x}} \sec^2 \sqrt{x}$; e. $x \cos x + \sin x$; f. $2x(\cos 2x - x \sin 2x)$; g. $-2 \cos x \sin x$; h. $3 \sin^2 x \cos x$; i. $-\frac{\cos x}{(1+\sin x)^2}$; j. $\frac{1}{1+\cos x}$; k. $-\frac{1}{1+\sin x}$; l. $-\frac{1}{(\cos x + \sin x)^2}$; m. $-\tan x$; n. $\sec^2 x e^{\tan x}$
3. a. The graphs are reflections of each other in the x -axis; b. The graphs are identical
4. a. $2 \cos 2x e^{\sin 2x}$; b. $2e^{2x} \cos(e^{2x})$; c. $\frac{2 \cos 4x}{\sqrt{\sin 4x}}$; d. $4 \cot 4x$; e. $9 \sin 3x (1 - \cos 3x)^2$; $2(\cos 2x \sin 4x + 2 \sin 2x \cos 4x)$; g. $-15 \cos^4 3x \sin 3x$; h. $\frac{20 \sin 5x}{(3+4 \cos 5x)^2}$; i. $15 \tan^2(5x-4) \sec^2(5x-4)$; j. $\frac{x \cos \sqrt{x^2+1}}{\sqrt{x^2+1}}$
6. a. $\frac{\pi x}{180}$; b. i. $\frac{\pi}{180} \cos x^\circ$, ii. $\frac{\pi}{180} \sec^2(x^\circ + 45^\circ)$, iii. $-\frac{\pi}{90} \sin 2x^\circ$
9. a. $\log_b P - \log_b Q$
10. b. $\frac{1}{2}((m+n) \cos(m+n)x + (m-n) \cos(m-n)x)$; b. $\frac{1}{2}(\cos(m+n)x + \cos(m-n)x) = \cos mx \cos nx$, $-\frac{1}{2}((m+n) \sin(m+n)x + (m-n) \sin(m-n)x)$
11. b. $\sin \left(\frac{n\pi}{2} + x \right)$